Exercise Sheet 1

The problems use complex finite-dimensional spaces. Extend each bracket table bilinearly, with zero diagonal and antisymmetry.

Exercise 1.1 (A two-dimensional algebra). Let $\mathfrak{g}$ have basis (h, e) and bracket $[h, e] = e$.

(i) Verify the Jacobi identity and compute $[ah + be, ch + de]$.

(ii) Find $Z(\mathfrak{g})$, $[\mathfrak{g}, \mathfrak{g}]$ and all ideals of $\mathfrak{g}$.

(iii) Identify $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$.

Exercise 1.2 (The trace quotient). Let $n \geq 2$. The matrix unit E_{ij} has entry 1 in position (i, j) and zero elsewhere.

(i) Prove that $\operatorname{tr}\colon \mathfrak{gl}_n(\mathbb{C}) \to \mathbb{C}$ is a Lie algebra homomorphism when $\mathbb{C}$ has zero bracket. Identify its kernel and the quotient of $\mathfrak{gl}_n(\mathbb{C})$ by this kernel.

(ii) Use matrix units to prove $[\mathfrak{gl}_n(\mathbb{C}), \mathfrak{gl}_n(\mathbb{C})] = \mathfrak{sl}_n(\mathbb{C})$.

Exercise 1.3 (The ideals of $\mathfrak{sl}_2$). Use the basis (h, e, f) of the lecture. Let $A = \operatorname{ad} h$.

(i) For $x = ah + be + cf$, find $(A^2 + 2A)x/8$ and $(A^2 - 2A)x/8$.

(ii) Prove that every nonzero ideal of $\mathfrak{sl}_2$ is $\mathfrak{sl}_2$. A nonabelian algebra with this property is called *simple*.

Exercise 1.4 (Irreducible Heisenberg modules). The Heisenberg algebra $\mathfrak{h}_3$ has basis (p, q, z) with $[p, q] = z$ and $[p, z] = [q, z] = 0$. Prove that every finite-dimensional irreducible complex $\mathfrak{h}_3$-module has dimension one. Describe these modules up to isomorphism.

Exercise 1.5 (Tensor products and duals). Let (V, ρ_V) and (W, ρ_W) be $\mathfrak{g}$-modules.

(i) Prove that the rule

$$\rho_{V\otimes W}(x) = \rho_V(x) \otimes \mathrm{id}_W + \mathrm{id}_V \otimes \rho_W(x)$$

defines a representation.

(ii) Prove that the algebraic dual V^* has a representation defined by

$$(\rho_{V^*}(x)\phi)(v) = -\phi(\rho_V(x)v).$$

If $D_V(x)$ is the matrix of $\rho_V(x)$ in a basis of V, find the matrix $D_{V^*}(x)$ of $\rho_{V^*}(x)$ in the dual basis.

Exercise 1.6 (Derivations of the Heisenberg algebra). Use the basis (p, q, z) of $\mathfrak{h}_3$ from Exercise 1.4. Find the matrix of every derivation in this basis. Which derivations are inner?

Exercise 1.7 (Complete reducibility for an abelian algebra). Let ρ be a representation of $\mathbb{C}t$ on V, and put $T = \rho(t)$. Prove that this module is completely reducible if and only if T is diagonalisable.